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Source: *Physiological Zoology*, Mar. - Apr., 1985, Vol. 58, No. 2 (Mar. - Apr., 1985), pp. 175-180

Published by: The University of Chicago Press. Sponsored by the Division of Comparative Physiology and Biochemistry, Society for Integrative and Comparative Biology

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BEHAVIORAL AND METABOLIC ADJUSTMENTS TO LOW TEMPERATURES IN THE LARGEMOUTH BASS (*MICROPTERUS SALMOIDES*)¹

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(Accepted 8/1/84)

Resting metabolic rate (MR), spontaneous swimming activity, and food intake were measured in order to learn the relationship between these responses and low acclimation temperatures. The fish were collected in the fall and placed in one of eight acclimation tanks ranging from 3 to 17 C at 2 C intervals for 3 wk. Food intake was measured during and after the acclimation period, and MR and activity were measured following the acclimation period. MR, measured in a closed system, was exponentially related to temperature with a Q_{10} of 3.0. Food intake declined linearly down to 10 C, below which it became negligible. Spontaneous swimming activity, measured in an annular tank, was similar for all fish at or above 7 C and was considerably reduced at 5 and 3 C. These data indicate that physiological systems respond very differently to low acclimation temperatures in the largemouth bass. MR shows no compensation. The central nervous system, which mediates activity cycles, shows perfect compensation from 17 to 7 C, with a decline at lower temperatures. An energy analysis reveals that food intake fails to meet even resting metabolic requirements at temperatures below 10 C and that the net cost of spontaneous swimming activity is a small portion of the total daily energy expenditure from 3 to 17 C. The data agree well both with observations on the effects of thermal acclimation on isolated tissues and with observations made on bass in the wild.

INTRODUCTION

Freshwater fish inhabiting the temperate zone experience annual cycles of temperature that have a major influence upon their physiology and behavior. Summer season water temperatures may average 20–35 C in ponds and lakes, whereas the winter temperature may be below 4 C for several months. During winter several factors may become limiting for fish, among which are food and oxygen supplies. Thermal compensation may impart some temperature independence in metabolism and activity, as occurs in salmonids such as the Atlantic salmon (Peterson and Anderson 1969). For many temperate species, including the brown bullhead, *Ictalurus nebulosus* (Crawshaw et al. 1982b) and the largemouth bass, *Micropterus salmoides* (Salamon 1979), activity is de-

pressed at low winter temperatures, a state that has been termed dormancy (see Crawshaw 1984). Several investigators have documented physiological adaptations that occur in ectotherms during winter dormancy. Greatly increased Q_{10} values have been observed at low temperatures in reptiles (Aleksiuk 1976), amphibians (Fitzpatrick, Bristol, and Stokes 1971; Packard 1972), and fish (Roberts 1964). In this paper, we investigate the nature of low-temperature dormancy in the largemouth bass, *Micropterus salmoides*. Whole-animal metabolism, spontaneous activity, and food intake were measured at temperature intervals between 3 and 18 C. The use of a very narrow temperature interval (2 C) allowed us to characterize accurately the functional relationships between the responses and the acclimation temperature. The results indicate very different patterns of acclimation for the three responses.

MATERIAL AND METHODS

Largemouth bass (*Micropterus salmoides*) ranging in size from 18 to 120 g were obtained in Oregon and Connecticut in the fall of the year and placed in groups

¹ This work was supported by National Science Foundation grant PCM 76-09658 and National Institutes of Health grant 1-R01-NS15318.

Physiol. Zool. 58(2):175–180. 1985.

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of six to eight in 110-liter temperature-controlled tanks. The temperature in the tanks was initially at 10 C and several days later was either raised or lowered 1 C·day⁻¹ in each tank until the final acclimation temperature for the tank was reached. Eight acclimation temperatures were established ranging from 3–17 C or 4–18 C (± 0.1 C) in increments of 2 C. The animals were considered acclimated after a minimum of 3 wk at a constant temperature. All the tanks were within an environmental room which was thermally, acoustically, and visually isolated. Each tank was covered by translucent plastic, and a 10L:14D light cycle was maintained throughout the study. Live goldfish were continuously available in the home tanks; the bass began eating within a week after transfer to the laboratory.

Oxygen consumption was measured with a portable closed circuit system that was placed into each acclimation tank. The system consisted of six components in series: a centrifugal pump, a disconnect point for the circuit, a flow indicator, a YSI O₂ electrode, a Plexiglas metabolic chamber, and a section of thin-walled stainless steel tubing for heat exchange. All components of the circuit were connected by Tygon tubing and were submerged in the tank water. Flow through the system was maintained between 500 and 800 ml·min⁻¹; the flow velocity in the metabolic chamber was 0.25–0.40 cm·s⁻¹. At this flow rate, adequate mixing occurred without forcing the fish to swim to maintain position. Baffles at the inlet and outlet of the metabolic chamber spread the flow across the diameter of the chamber. The O₂ electrode was connected to a YSI Model 54 Dissolved Oxygen meter, the output of which was recorded on a Honeywell strip chart recorder.

To begin an experiment, the respirometer was lowered into an acclimation tank with the entrance to the metabolic chamber open. A fish was then directed into the chamber, which was then sealed. The pump was turned on, drawing water from the tank (circuit open). The tank was then covered and the fish left in the chamber overnight. After 14–18 h, a small lid in the tank cover was opened and air bubbles in the metabolic chamber were gently

purged. The fish was then left undisturbed for 0.5–1 h. Next, the system was closed and the first measurement period begun. The PO₂ was continually monitored and allowed to fall from about 150 torr to 90–110 torr, at which time the circuit was opened and aerated water was again drawn by the pump from the acclimation tank (open circuit). Fish observed through a small slit in the cover never appeared disturbed by the fall in PO₂. Twenty to 30 min were allowed to elapse between measurement periods, and five to six measurements were made per day, with the lowest rate of O₂ uptake usually occurring near the end of the day. We utilized the lowest value so obtained as representative of a resting value for that fish. The O₂ electrode was calibrated, using the Winkler technique (Hoar and Hickman 1975), with water samples drawn from the circuit outflow immediately prior to closing the system at the beginning of a trial and immediately following reopening the circuit at the end of the trial. Below 9 C the probe was found to account for a significant proportion of the oxygen consumed. This proportion was subtracted from the total. Metabolic rate was calculated as in Crawshaw et al. (1982a).

Some fish were left in the metabolic chamber a second night for the purpose of determining whether an additional 24 h in the chamber would further reduce metabolic rate. Since this additional time in the chamber produced only a small additional fall in MR (about 5%), all measurements were made 14–18 h after placement in the chamber. Fish were removed 24 h after they had been placed in the metabolic chamber and were anesthetized with MS222, weighed, measured for length, tagged with colored thread, and returned to the acclimation tank.

Activity was measured in a portable annular tank that was placed atop the acclimation tank holding the fish to be tested. The inside radius of the annular tank was 39.4 cm, and the annulus width was 8.0 cm. If one assumes that the fish swam in the middle of the annulus, one circuit equaled 2.72 m. Water was drawn from the acclimation tank by a centrifugal pump and flowed into the annulus from a T connector that directed the water flow

in two opposite paths. The water exited the annulus 180° from its point of entry and flowed back into the acclimation tank. There was, thus, no net directional flow around the annulus. The temperature in the annular tank was within 0.5 C of the acclimation tank temperature. For the measurement of spontaneous swimming activity, termed "routine activity" by Beamish (1979), three beams of red light at different heights were directed at photocells across the annulus. Each time the fish interrupted one of the beams, an event was recorded on an Esterline-Angus recorder. The behavior of the fish was not noticeably affected by the light beams.

To begin a trial, a fish was placed in the annulus, which was subsequently covered with translucent plastic. The fish was then left undisturbed for 50–72 h and activity monitored continuously under the same 10L:14D photoperiod. After the trial, the fish was removed from the annulus, anesthetized, weighed, measured, and marked (if this had not been previously done).

A separate group of fish was used for measurement of food intake. These animals were kept at acclimation temperatures from 4 to 18 C. They were fed mealworms each afternoon; uneaten worms were removed the following mid-

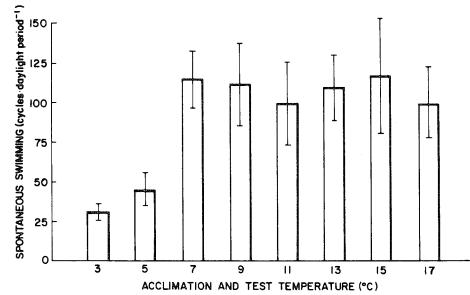


FIG. 2.—Spontaneous swimming activity around an annular tank. Data are for the daylight period of the second day the fish was in the apparatus. Values given are means $\pm$ 1 SEM. The number of fish in each acclimation group from 3 to 17 C, respectively, was six, seven, nine, seven, nine, seven, five, and eight.

morning. After all groups reached the acclimation temperature, the number of worms eaten daily was recorded for a period of 4 wk. The number of worms fed was always greater than the number eaten—except for the 18 C acclimation group, which ate all the worms fed to it during the fourth week.

RESULTS

Metabolic measurements at different temperatures showed no major discontinuities. These data were therefore treated as a continuous function (fig. 1). Curve fitting using the method of least squares indicated that the data were best described by an exponential equation (coefficient of determination = $r^2 = .963$), which is $y = 4.48 e^{0.11x}$; $y = \text{ml O}_2 \cdot \text{kg}^{-1} \cdot \text{h}^{-0.77}$; $x = \text{degrees C}$. This relationship has a Q_{10} of 3.0 and is depicted by the dashed line (fig. 1).

Routine diurnal activity in the annular tank is shown in figure 2. Fish placed in the tank went into a period of exploratory activity which lasted up to 6 h (the data from the first day were not included in the analysis). Thereafter, swimming around the tank became more regular, and periods of quiescence as well as continuous swimming occurred. Most activity occurred during the daylight. Routine diurnal activity was independent of temperature down to 7 C (least-squares regression analysis: $r^2 = .139$ [$P < .1$]). A major discontinuity occurred below 7 C, with

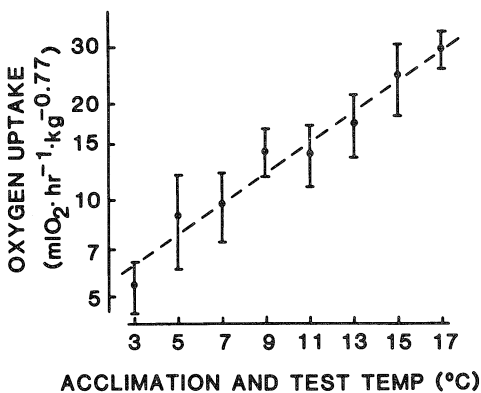


FIG. 1.—A semilogarithmic plot of the resting metabolic rate at eight acclimation temperatures. Data were obtained in a closed-system respirometer and represent the lowest value obtained for each individual. Values are shown $\pm$ 2 SEM. The number of fish in each acclimation group from 3 to 17 C, respectively, were six, five, seven, 10, 10, four, four, and four. The dashed line represents the least-squares best fit ($Q_{10} = 3.0$).

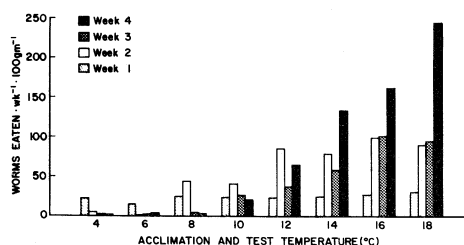


FIG. 3.—Food intake of bass from eight acclimation temperatures over a 4-wk period. The number of individuals in each acclimation group from 4 to 18 C, respectively, was five, five, six, six, six, five, five, and five.

spontaneous swimming activity much lower at 5 and 3 C.

Food intake (fig. 3) was nearly the same for all groups during the first week of acclimation but fell during the third week for all three groups below 10 C. This contrasted with the 10 C acclimation group whose food intake was constant and with all groups above 10 C whose food consumption rose over the 4 wk. During the fourth week, the 4 C group ate nothing, and little food was consumed by the 6 and 8 C groups. Food intake for the fourth week was analyzed statistically. Since the relationship between acclimation temperature and food intake did not appear to be continuous, a break point regression analysis was performed as outlined by Withers (1980). First, the regression for the lowest three temperatures was computed, and then values for each successively higher acclimation temperature were added and the regression recalculated. Next, the data for the highest three temperatures were analyzed, the values for each successively lower acclimation temperature were added, and the regression was recalculated. The coefficients of determination (r^2) obtained for each combination were compared; this approach indicated that the best fit was obtained with linear regressions for the data above 9 C ($y = 26.6x - 245.5$; $r^2 = .975$) and below 9 C ($y = 0.9x - 3.0$; $r^2 = .793$); y = worms eaten $\cdot$ wk $^{-1}$ $\cdot$ 100 g $^{-1}$; x = degrees C.

For a comparison of their interrelationships, the fitted curves for metabolic rate, activity, and food intake were converted to units of energy for a standardized 70-g fish (fig. 4). To accomplish this, the following data were utilized: (1) One liter O $_2$

utilized = 19.40 kJ (Brett and Groves 1979). (2) The net cost of swimming at 1 body length $\cdot$ s $^{-1}$ was calculated from the data of Beamish (1970). Beamish found that the logarithm of active oxygen consumption of largemouth bass increased linearly with weight for different temperatures, with a slope near unity. In his figure 5, oxygen consumption is plotted against swimming speed for a 150-g fish at 10, 15, and 20 C. The data at these temperatures were utilized, for 0 and 1 body length $\cdot$ s $^{-1}$, to plot a linear relationship between the logarithm of the metabolic rate of a 70-g fish (metabolic rate of the 150-g fish $\cdot$ 70/150) and the temperature. This plot was employed to obtain the portion of the total metabolic rate during activity that was attributable to swimming (rate at 1 body length $\cdot$ s $^{-1}$ less rate at 0 body length $\cdot$ s $^{-1}$). Since the distance around the annular tank was known, by taking 17.5 cm as the length of the standardized 70-g fish and assuming a swimming speed of 1 body length $\cdot$ s $^{-1}$, one could calculate the daily time spent swimming at each temperature. The product of the swimming time and the net metabolic rate gave the net daily energy cost for swimming. (3) Digestible energy (utilizable food intake) was obtained by assuming mealworms weighed 100 mg, were 61% water, contained 23.88 kJ $\cdot$ g $^{-1}$ dry weight (Avery 1971), and were assimilated with an efficiency of 83.2%—a value given by Brett and Groves (1979) for

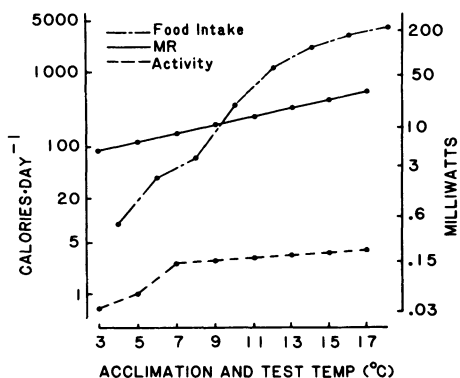


FIG. 4.—A semilogarithmic plot comparing food intake, resting metabolic rate, and spontaneous swimming activity in terms of energy made available to or utilized by a standardized 70-g bass at temperatures between 3 and 18 C.

carnivorous fish feeding on invertebrates with a hard exoskeleton. The assimilation efficiency was corrected for temperature by utilizing the data of Elliott (1976). Two major points emerged when all the dependent variables were converted to units of energy. First, spontaneous diurnal activity formed a small portion of daily energy expenditure. Second, as the acclimation temperature fell from 12 to 8 C, digestible energy declined from an adequate surplus to a level of negative energy balance.

DISCUSSION

The depression of activity below 7 C confirmed that the bass were in a dormant state at these low temperatures. The nearly perfect compensation seen from 7 to 17 C (type 2—Precht [1958]) indicates a high degree of acclimation of central nervous system function. A similar response was seen in the brown bullhead (Crawshaw et al. 1982b). As noted by Prosser and Nelson (1981), the poikilotherm brain is highly aerobic in its metabolism, and the enzyme systems in these animals often show marked thermal acclimation. Muscle and liver metabolisms, on the other hand, frequently show little thermal acclimation.

These latter tissues account for a significant proportion of the total metabolic rate; in this study, the resting metabolic rate likewise showed no compensation (Precht type 4) over the range of temperatures studied. Although a metabolic "shutdown" (Precht type 5) as seen in the sunfish, *Lipomus gibbosus* (Roberts 1964), would produce the largest metabolic declines during dormancy, the type 4 response ($Q_{10} = 3.0$) exhibited by bass also leads to quite low rates of energy consumption at low temperatures. The metabolic rates determined in this study were, at overlapping temperatures, similar to those reported by Venables, Pearson, and Fitzpatrick (1977). The results of these

two studies, taken together, indicate that the resting metabolic rate of the largemouth bass increases in a monotonic exponential manner from 3 to 30 C.

Food intake, like spontaneous swimming activity, showed a definite discontinuity. Since food intake declined to very low levels at temperatures 3–5 C above those at which the activity declined, the effect cannot be attributed to a generalized decrease in central nervous system activity. Similarly, Salamon (1979) observed that wild overwintering bass foraged only when their liver glycogen was severely depleted.

The energetic analysis of the variables measured indicates an adequate energy surplus down to 12 C. At 10 C there is little energy scope, and at lower temperatures food intake is insufficient to cover even the resting metabolic rate. The temperatures at which feeding declined rapidly correspond to fall lake temperatures when wild bass schooled, dispersed, and sought out secluded areas on the lake bottom (Salamon 1979). Since the fish were active at the higher temperatures for only about $29 \text{ min} \cdot \text{day}^{-1}$, the daily net energy expenditure for spontaneous swimming activity was quite low.

In this study, by utilizing closely spaced acclimation temperatures, we have shown that low temperatures can produce very different patterns of acclimation for different physiological systems. Resting metabolic rate follows a monotonic, exponential decrease with lowered temperature. Food intake (plotted as energy assimilated) remains well above resting metabolic requirements down to 12 C and then declines rapidly. Spontaneous activity shows almost perfect compensation down to 7 C, after which there is a marked dropoff. We interpret the abrupt decreases in food intake and spontaneous activity as important aspects of low-temperature dormancy in largemouth bass.

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